Assignment 2 :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>KBTCOE Connect — Semantic Campus Portal</title>
</head>
<body>

  <!-- Top Page Header -->
  <header>
    <div>
      <img src="kbtcoe-logo.svg" alt="Official crest and banner of MVP Samaj's
KBTCOE Connect" width="480" height="100">
      <p>
        <strong>MVP Samaj's Karmaveer Adv. Baburao Ganpatrao Thakare College of
Engineering</strong><br>
        <em>Affiliated to Savitribai Phule Pune University (SPPU) | Approved by
AICTE, New Delhi</em>
      </p>
    </div>
    <hr>

    <!-- Semantic Primary Navigation -->
    <nav aria-label="Main Portal Navigation">
      <ul>
        <li><a href="#student-profile">Student Profile</a></li>
        <li><a href="#technical-blogs">Technical Blog Articles</a></li>
        <li><a href="#achievements-competitions">Achievements &amp;
Hackathons</a></li>
        <li><a href="#opportunities-bulletin">Opportunities Bulletin</a></li>
        <li><a href="#campus-announcements">Campus Notices</a></li>
      </ul>
      <hr>
    </nav>
  </header>

  <!-- Main Content Region -->
  <main>

    <!-- Section 1: Student Profile Showcase -->
    <section id="student-profile">
      <header>
        <h1>Student Profile Spotlight</h1>
        <p>Verified academic and professional credentials from the Department of
Information Technology.</p>
      </header>

      <article>
        <header>
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          <h2>Snehal Rajesh Deshmukh</h2>
          <p>
            <strong>PRN:</strong> <code>72183944C</code> |
            <strong>Branch:</strong> Information Technology (Final Year B.E.) |
            <strong>Anticipated Graduation:</strong> <time
datetime="2026-06">June 2026</time>
          </p>
        </header>

        <section>
          <h3>Professional Summary</h3>
          <p>
            Final-year undergraduate student specializing in enterprise backend
architecture, database optimization, and cloud services. Technical lead of the
KBTCOE ACM Student Chapter, mentor for junior coding bootcamps, and passionate
open-source contributor.
          </p>
        </section>

        <section>
          <h3>Core Competencies &amp;amp; Technical Skills</h3>
          <ul>
            <li><strong>Programming Languages:</strong> Java (Core &amp;amp;
Advanced), JavaScript (ES6+), Python, SQL</li>
            <li><strong>Frameworks &amp;amp; Web Technologies:</strong> Spring Boot,
Node.js, Express.js, React.js, HTML5, CSS3</li>
            <li><strong>Databases:</strong> MySQL 8, PostgreSQL, Redis
(Caching)</li>
            <li><strong>Engineering Practices:</strong> RESTful API Design, Git
Version Control, Microservices, CI/CD Pipelines</li>
          </ul>
        </section>

        <section>
          <h3>Academic &amp;amp; Portfolio Links</h3>
          <ul>
            <li><a href="https://github.com" target="_blank" rel="noopener
noreferrer">Visit Snehal's GitHub Profile to inspect open source
repositories</a></li>
            <li><a href="https://linkedin.com" target="_blank" rel="noopener
noreferrer">Connect with Snehal on LinkedIn for professional
correspondence</a></li>
            <li><a href="mailto:snehal.deshmukh@kbtcoe.org">Send academic email
to snehal.deshmukh@kbtcoe.org</a></li>
          </ul>
        </section>

        <footer>
          <p>
            <small>Profile status: <strong>Verified Student Member</strong> |
Last updated on <time datetime="2026-10-10">10 October 2026</time></small>
          </p>
        </footer>
      </article>
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      <hr>
    </section>

    <!-- Section 2: Technical Blog Articles -->
    <section id="technical-blogs">
      <header>
        <h1>Technical Blog Articles</h1>
        <p>Peer-reviewed articles, engineering insights, and development guides
authored by KBTCOE students.</p>
      </header>

      <!-- Blog Article 1 -->
      <article>
        <header>
          <h2>Architecting Scalable REST APIs for Campus Portals using Node.js
and MySQL</h2>
          <p>
            Published by <strong>Snehal Deshmukh</strong> on
            <time datetime="2026-10-12">12 October 2026</time> &bull;
            <em>Reading Time: 6 minutes</em>
          </p>
        </header>

        <section>
          <h3>Introduction and Architectural Problem Statement</h3>
          <p>
            Modern campus portals manage fluctuating traffic surges during
semester registration and placement season. Traditional monolithic architectures
often fail under sudden concurrent loads due to database connection exhaustion
and blocking I/O operations. In this technical writeup, we analyze how a
decoupled 3-tier architecture with connection pooling resolves these
bottlenecks.
          </p>

          <figure>
            <img src="rest-architecture.svg" alt="Three-tier REST API
architecture showing Client Tier with React SPA, REST API Tier with Node.js
Express, and Database Tier with MySQL" width="600" height="220">
            <figcaption>Figure 2.1: Data flow and connection pool architecture
in KBTCOE Connect.</figcaption>
          </figure>

          <h3>Key Architectural Principles</h3>
          <ol>
            <li>
              <strong>Connection Pooling:</strong> Utilizing
<code>mysql2/promise</code> connection pools prevents the overhead of creating
and tearing down TCP handshakes on every incoming HTTP request.
            </li>
            <li>
              <strong>Parameterized Queries:</strong> Protecting campus
databases from SQL injection vulnerabilities by enforcing prepared statements
across all student CRUD operations.
            </li>
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            <li>
              <strong>Stateless Authentication:</strong> Employing JSON Web
Tokens (JWT) to verify student identities across distributed endpoints without
server-side memory session bloat.
            </li>
          </ol>
        </section>

        <footer>
          <p>
            <strong>Categorized under:</strong> System Architecture, Node.js,
Express, MySQL, Security<br>
            <a href="#student-profile">Read more publications by this author</a>
          </p>
        </footer>
      </article>

      <hr>

      <!-- Blog Article 2 -->
      <article>
        <header>
          <h2>Demystifying Git Branching Strategies for College Hackathon
Teams</h2>
          <p>
            Published by <strong>Rohit Sandip Patil</strong> on
            <time datetime="2026-09-28">28 September 2026</time> &bull;
            <em>Reading Time: 4 minutes</em>
          </p>
        </header>

        <section>
          <h3>The Chaos of Rapid Hackathon Collaboration</h3>
          <p>
            During 36-hour hackathons like Smart India Hackathon, teams
frequently stumble into merge conflicts, overwritten code, and broken master
branches. Adopting a structured, lightweight Git strategy transforms team
productivity.
          </p>
          <p>
            We recommend a three-branch workflow:
          </p>
          <ul>
            <li><strong><code>main</code>:</strong> Production-ready code; demo
deployable at any moment.</li>
            <li><strong><code>dev</code>:</strong> Integration branch where team
members merge verified features.</li>
            <li><strong><code>feature/*</code>:</strong> Isolated branches for
individual sub-modules (e.g., <code>feature/login-ui</code>,
<code>feature/api-routes</code>).</li>
          </ul>
        </section>

        <footer>
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          <p>
            <strong>Categorized under:</strong> Git, DevOps, Teamwork,
Hackathons<br>
            <a href="mailto:rohit.patil@kbtcoe.org">Discuss this article with
the author</a>
          </p>
        </footer>
      </article>
      <hr>
    </section>

    <!-- Section 3: Achievements & Competitions -->
    <section id="achievements-competitions">
      <header>
        <h1>Student Achievements &amp;amp; Campus Competitions</h1>
        <p>Documenting institutional accolades, national hackathon milestones,
and upcoming technical events.</p>
      </header>

      <!-- Achievement Article -->
      <article>
        <header>
          <h2>National Milestone: Smart India Hackathon (SIH) 2025 Regional
Finalist</h2>
          <p>
            Conferred on: <time datetime="2025-12-18">18 December 2025</time> |
            Issued by: <strong>Ministry of Education's Innovation Cell, Govt. of
India</strong>
          </p>
        </header>
        <p>
          A multidisciplinary squad representing KBTCOE (comprising Computer,
IT, and Mechanical students) developed an automated rooftop solar panel
fault-detection system using drone thermal imagery and edge computing
algorithms.
        </p>
        <ul>
          <li><strong>Team Name:</strong> Team ByteCraft KBTCOE</li>
          <li><strong>Problem Statement ID:</strong> SIH1422 (Renewable
Energy)</li>
          <li><strong>Key Result:</strong> Recognized as Regional Runner-Up with
a cash citation of INR 25,000.</li>
        </ul>
      </article>

      <!-- Competition Announcement Article -->
      <article>
        <header>
          <h2>Upcoming Event: KBTCOE CodeSprint 2026 &amp;mdash; Annual Algorithmic
Challenge</h2>
          <p>
            Event Date: <time datetime="2026-11-10">10 November 2026</time> |
            Organized by: <strong>Computer Engineering Students' Association
(CESA)</strong>
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          </p>
        </header>
        <p>
          CodeSprint 2026 is an individual 3-hour competitive programming
contest designed to prepare second, third, and final year students for technical
recruitment rounds of Tier-1 product companies.
        </p>
        <ul>
          <li><strong>Eligibility:</strong> Open to all enrolled undergraduate
engineering students of KBTCOE.</li>
          <li><strong>Topics:</strong> Data Structures, Dynamic Programming,
Graph Theory, and Combinatorics.</li>
          <li><strong>Registration Deadline:</strong> <time
datetime="2026-11-05">05 November 2026</time></li>
        </ul>
        <p>
          <a
href="mailto:cesa@kbtcoe.org?subject=CodeSprint%202026%20Registration%20Inquiry"
>Contact CESA Student Coordinators to Register</a>
        </p>
      </article>
      <hr>
    </section>

    <!-- Section 4: Opportunities Bulletin -->
    <section id="opportunities-bulletin">
      <header>
        <h1>Job &amp;amp; Internship Opportunities Bulletin</h1>
        <p>
          <em>Notice: Sample demonstration records curated for the KBTCOE
Connect campus platform.</em>
        </p>
      </header>

      <table border="1" cellpadding="8" cellspacing="0" width="100%">
        <caption><strong>Table 2.1: Semantic Catalog of Verified Campus
Opportunities</strong></caption>
        <thead>
          <tr bgcolor="#E2E8F0">
            <th scope="col">Listing ID</th>
            <th scope="col">Position / Title</th>
            <th scope="col">Organization</th>
            <th scope="col">Eligible Branches</th>
            <th scope="col">Location</th>
            <th scope="col">Deadline</th>
            <th scope="col">Application Action</th>
          </tr>
        </thead>
        <tbody>
          <tr>
            <td align="center"><strong>JOB-201</strong></td>
            <td>Graduate Trainee Engineer &amp;mdash; Software</td>
            <td>Emerson Innovation Center</td>
            <td>Computer, IT, Instrumentation</td>
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            <td>Nashik MIDC</td>
            <td><time datetime="2026-11-25">25 November 2026</time></td>
            <td><a href="https://www.emerson.com" target="_blank" rel="noopener
noreferrer">Apply on Emerson Careers Portal</a></td>
          </tr>
          <tr>
            <td align="center"><strong>INT-202</strong></td>
            <td>Cloud Infrastructure Intern</td>
            <td>Infosys Springboard Campus Program</td>
            <td>All Disciplines (TE, BE)</td>
            <td>Virtual / Hybrid</td>
            <td><time datetime="2026-12-01">01 December 2026</time></td>
            <td><a href="https://infyspringboard.onwingspan.com" target="_blank"
rel="noopener noreferrer">Register via Infosys Springboard</a></td>
          </tr>
          <tr>
            <td align="center"><strong>CMP-203</strong></td>
            <td>TCS CodeVita Season 14 Global Contest</td>
            <td>Tata Consultancy Services</td>
            <td>All Engineering Branches</td>
            <td>Online Contest</td>
            <td><time datetime="2026-12-15">15 December 2026</time></td>
            <td><a href="https://tcscodevita.com" target="_blank" rel="noopener
noreferrer">Register for TCS CodeVita Season 14</a></td>
          </tr>
        </tbody>
        <tfoot>
          <tr bgcolor="#E2E8F0">
            <th colspan="5" align="left">
              Total Verified Opportunities Listed: 3 Demonstration Records
            </th>
            <th colspan="2" align="center">
              Coordinated by: <strong>KBTCOE T&amp;amp;P Cell</strong>
            </th>
          </tr>
        </tfoot>
      </table>
      <hr>
    </section>

  </main>

  <!-- Semantic Aside for Tangential Information -->
  <aside id="campus-announcements" aria-label="Campus Notices and Guidelines">
    <header>
      <h2>Campus Notices &amp;amp; Student Guidelines</h2>
      <p>Important administrative updates for the current academic session.</p>
    </header>

    <section>
      <h3>University Examination Notices (SPPU)</h3>
      <p>
        Savitribai Phule Pune University has published the tentative timetable
for the
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        <strong>Semester VII &amp;amp; V In-Semester Examinations</strong>.
      </p>
      <p>
        Commencement Date: <time datetime="2026-11-02">02 November
2026</time><br>
        Detailed schedules are posted on the departmental display boards.
      </p>
    </section>

    <section>
      <h3>KBTCOE Connect Community Guidelines</h3>
      <ol>
        <li>Articles must be original, technical, and adhere to academic
anti-plagiarism standards.</li>
        <li>Opportunity postings must clearly state eligibility, stipend/salary,
and official contact emails.</li>
        <li>Treat fellow students, faculty, and alumni with professional
courtesy.</li>
      </ol>
    </section>

    <section>
      <h3>Trending Campus Tags</h3>
      <p>
        <code>#ReactJS</code> &bull;
        <code>#NodeJS</code> &bull;
        <code>#SmartIndiaHackathon</code> &bull;
        <code>#SPPUExams</code> &bull;
        <code>#CampusPlacements2026</code>
      </p>
    </section>
  </aside>

  <hr>

  <!-- Semantic Page Footer -->
  <footer>
    <section>
      <p>
        <strong>KBTCOE Connect &mdash; The Professional Network for KBTCOE
Students</strong><br>
        MVP Samaj's Karmaveer Adv. Baburao Ganpatrao Thakare College of
Engineering<br>
        Udoji Maratha Boarding Campus, Gangapur Road, Nashik &mdash; 422013,
Maharashtra, India.
      </p>
      <p>
        <strong>Helpdesk Email:</strong> <a
href="mailto:connect.support@kbtcoe.org">connect.support@kbtcoe.org</a> |
        <strong>T&amp;amp;P Office:</strong> <a
href="mailto:tnp@kbtcoe.org">tnp@kbtcoe.org</a> |
        <strong>Phone:</strong> +91 (0253) 2571439
      </p>
      <p>
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        <a href="https://kbtcoe.org" target="_blank" rel="noopener
noreferrer">Visit Official KBTCOE Institution Portal</a> &bull;
        <a href="http://www.unipune.ac.in" target="_blank" rel="noopener
noreferrer">Visit Savitribai Phule Pune University (SPPU) Portal</a> &bull;
        <a href="#student-profile">Back to Top of Page &uarr;</a>
      </p>
    </section>
    <hr>
    <p align="center">
      <small>
        &copy; <time datetime="2026">2026</time> KBTCOE Connect. Academic
Practical Deliverable &mdash; Department of Information Technology &amp;
Computer Engineering.<br>
        All synthetic student profiles and opportunity records are intended for
academic demonstration.
      </small>
    </p>
  </footer>

</body>
</html>
```